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Coffee: Grounds for concern?

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Do you find yourself looking forward to that first cup of coffee in the morning? How about the next cup? If it's part of your routine to drink coffee on a regular basis, you're not alone.

You also have company when it comes to concerns about the health effects of coffee - hundreds of studies have addressed coffee's effect on the body and whether or not it causes harm. Perhaps you've heard it's good to drink coffee when you have a headache, or bad to drink it if you have stomach problems. So, what is myth, and what is fact?

Coffee's punch

The effects of coffee on the body fall into several categories:

- Stimulant effects. Because one of its major ingredients is caffeine, coffee is a well-established stimulant, meaning that it stimulates the nervous system, including such diverse networks as the nerves controlling intestinal activity, blood pressure, and airway size. As a result, any caffeine-containing food or beverage (including tea, cola, and chocolate milk) may impair sleep, but avoiding coffee late in the day is usually an easy way to avoid this problem. Jitters and anxiety may also be related to caffeine's stimulant effects.
- Heartburn. Even decaffeinated coffee can stimulate secretion of stomach acid, leading to heartburn.
- Diuretic features. Caffeine is also a diuretic - it encourages the kidneys to produce more urine; together with the water contained in coffee, some people experience an annoying increase in urination.

Coffee can have other effects on the body. For example, yellowed teeth are common among regular coffee drinkers. And then there are the occasional injuries related to burns from hot coffee.

As for the overall health risks of coffee or caffeine use, concerns have been raised by studies over the past 50 years, including an association with stomach problems, pancreatic and bladder cancer, fibrocystic breast disease, and gallbladder disease, among other conditions. When rigorously analyzed, these studies fall far short of implicating modest coffee consumption as a significant health risk for most people.

One largely discredited study that found that coffee intake was associated with an increased risk of pancreatic cancer is often used as a model for how a flawed study design can lead to misleading research results. In that study, researchers analyzed a number of "exposures" among patients with pancreatic cancer, including coffee intake. Because of the number of factors examined, many scientific researchers call this sort of study a "fishing expedition."

As reasonable as it may seem to examine many factors at once, the problem is that if you look at enough exposures, one or more will show an association just by chance. So, there is a danger of generating misleading results if you cast too wide a net, and well-respected researchers avoid doing so, or they make statistical adjustments to account for analyzing many variables in a single study.

Caffeine may actually be good for you

It may surprise you to learn that there are therapeutic uses of caffeine other than as a stimulant to avoid sleep (as in NoDoz, Vivarin, and many others):

- Newborns, especially those who are premature or have undergone surgery just after birth, may be treated with caffeine to stimulate their breathing.
- Many over-the-counter headache or pain remedies include caffeine (such as Excedrin, which also contains acetaminophen and aspirin). The effectiveness of these agents may be related, at least in part, to the treatment of caffeine withdrawal, a common cause of headaches.
- Studies have examined whether caffeine could be useful in the treatment of asthma, given its dilating effects on airways, and several have found modest benefits. In fact, some recommend that coffee intake be avoided before breathing tests because it might mask the abnormalities the breathing tests are looking for.
- Coffee has several effects on the intestinal tract. Although it increases stomach acid, it has not been definitively linked to ulcer disease. And it may even serve as a remedy for constipation.
- There is some evidence that coffee consumption may reduce the risk of colon cancer, although this association has not been widely accepted and has not led to any specific recommendations to encourage coffee drinking.

- The results of several studies suggest that coffee reduces the risk of Parkinson's disease. Some research suggests that as little as one cup a day can reduce the risk by 30%. And, among people who already have Parkinson's disease, coffee may reduce symptoms and worsening of disease over time.
- There is evidence that coffee consumption might reduce the risk of diabetes, Alzheimer's disease, kidney stones, gallstones, and depression.
- Some studies suggest a link between coffee consumption and a reduced risk of cirrhosis and gout.

However, contrary to popular belief, coffee is not an effective way to reverse the effects of inebriation.

For the vast majority of coffee drinkers, the news is encouraging: the health risks are minimal, if present at all. There are probably rare, high-risk patients who are better off avoiding the stimulant action of caffeine or the heartburn provoked even by decaffeinated coffee.

Perhaps the biggest problem of all for most coffee drinkers is its financial impact: while it may cost pennies a day when made at home, the cost of coffee at coffee shops can add up fast. In fact, the best reasons to drink coffee or to avoid it may have little to do with your health.

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